EXISTENCE OF PERIODIC SOLUTIONS TO EQUATIONS WITH MEASURES VIA THE SHOOTING METHOD

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Abstract. In this paper, we investigate the existence of periodic solutions for systems of measure differential equations (MDEs). Our approach is based on the shooting method, which allows us to establish a theoretical result guaranteeing the existence of solutions to the problem under consideration. Moreover, the shooting method serves as a robust tool for the numerical approximation of such systems. In the final section, we apply this approach to demonstrate the existence of periodic solutions for a forced pendulum subject to impulsive forces. Additionally, we present a computational implementation to obtain approximate solutions.

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1. Introduction and Main Theorem

The goal of this paper is to establish the existence of periodic solutions to *Measure Differential Equations* (briefly, MDEs). More specifically, we seek solutions to the following boundary value problem

$$\begin{cases} d\varphi = F(t, \varphi(t))\, d\nu \\ \varphi(0) - \varphi(T) = 0, \end{cases} \tag{1.1}$$

where $\nu : \mathscr{B}([0,T]) \to \mathbb{R}^m$ is a vector measure (see [9]) with finite total variation, $\mathscr{B}([0,T])$ is the σ-algebra of all Borel measurable subsets of $[0,T]$, and $F : [0,T] \times \mathbb{R}^n \to \mathbb{R}^{n\times m}$ is a matrix-valued Carathéodory function with respect to ν. That is, $F(t,x)$ is Borel measurable as a function of t for every $x \in \mathbb{R}^{n\times m}$, and continuous as a function of x for $|\nu|$-a.e. $t \in [0,T]$.

In what follows, we denote by $BV([0,T],\mathbb{R}^n)$ the set of all $\mathbb{R}^n$-valued functions of bounded variation. If $\varphi \in BV([0,T],\mathbb{R}^n)$, we denote by μ_φ the Lebesgue–Stieltjes measure induced by φ (see [7, 10, 11, 15, 24]). Whenever $f \in L^1([0,T],\mu_\varphi)$, for simplicity we write $\int f\, d\varphi$ instead of $\int f\, d\mu_\varphi$.

In the literature, several notions of solutions for MDEs have been proposed, which in some cases lead to non-equivalent results for the same equation (see [29]). In this paper, we adopt the definition introduced in [4].

Definition 1.1 (see [4])**.** *We say that $\varphi \in BV([0,T],\mathbb{R}^n)$ is a solution of $d\varphi = F(t,\varphi(t))\, d\nu$ if, for every $t \in [0,T]$, $\varphi(t)$ is left-continuous and, for*

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every Borel measurable set A, the following holds:

$$\int_A d\varphi = \int_A F(s, \varphi(s))\, d\nu. \tag{1.2}$$

Many of the MDEs considered in the literature are particular instances of (1.1). For example, it is common to encounter systems of MDEs written in the form

$$d\varphi = f(t, \varphi) + g(t, \varphi) d\mu,$$

where μ is a real-valued signed Borel measure and f and g are vector-valued functions (see [6]). Clearly, this equation can be viewed as a particular case of (1.1) by taking $F = (f, g)$ and $\nu = (\lambda, \mu)^t$, where λ denotes the Lebesgue measure on $\mathbb{R}$.

The Radon–Nikodym theorem allows us to rewrite equations involving vector measures in terms of positive measures. Let ν be a vector measure with values in $\mathbb{R}^m$. Then there exists a function $H \in L^1(|\nu|; \mathbb{R}^m)$ such that, for every measurable set E, the following representation holds:

$$\nu(E) = \int_E H(s)\, d|\nu|.$$

As a consequence, any solution of (1.1) is also a solution of

$$(MDE) \qquad \begin{cases} d\varphi = f(t, \varphi(t))\, d\mu \\ \varphi(0) - \varphi(T) = 0, \end{cases}$$

where $f : [0, T] \times \mathbb{R}^n \to \mathbb{R}^n$ is the vector field defined by $f = FH$, and μ is a positive measure. Accordingly, throughout this paper we focus our analysis on problem (MDE). Depending on the measure ν, the function f may be discontinuous with respect to time, even if F is continuous with respect to that variable. This observation highlights the importance of not imposing, or imposing only in the weakest possible sense, continuity assumptions on f with respect to t.

MDEs appear in nonsmooth models across several branches of science, ranging from the motion of molecules in ideal gases to the dynamics of impacts between a ball and a racket. In physics and mechanical engineering, MDEs arise in the study of granular matter and sandpiles; in robotics, they are used to model impacts at joints (see [6] and [19]). A particular case in which MDEs arise is given by impulsive models; see [2, 3, 18].

For example, in mechanical systems, the impact between two bodies can be regarded as a short-duration phenomenon that produces a sudden change in the system dynamics. Consequently, impacts are usually represented as large forces acting over an infinitesimally short time interval. Control theory provides another source of equations involving measures. For instance, in the optimization of space flight mechanics and in robotic motion, classical variational methods do not allow one to obtain optimal controls with an impulsive character (see [29]).

Beyond applications, differential equations with measures constitute, from a purely mathematical point of view, a natural generalization of ordinary differential equations, and, in our opinion, this fact justifies their study.

The problem of finding periodic solutions to impulsive systems has attracted considerable attention. It has been addressed in the literature using

various techniques; see, for example, [8, 13, 14, 16, 20–23, 27, 28, 30]. To the best of our knowledge, the present paper is the first to approach this problem via the shooting method.

Several integration theories have been used to deal with problems such as those described above, namely the Lebesgue–Stieltjes, Perron–Stieltjes, and Kurzweil–Henstock integrals (see, for example, [5, 17, 25, 26]).

We employ the shooting method to find solutions to the periodic boundary value problem associated with a first-order system of differential equations. The idea consists of seeking a solution u_α to the initial value problem

$$(IVP) \qquad \begin{cases} u' = f(t, u(t))d\mu \\ u(0) = \alpha, \end{cases}$$

and then finding a value $\alpha \in \mathbb{R}^2$ such that $u_\alpha(T) = \alpha$. That is, α is a fixed point of the Poincaré operator $P : \mathbb{R}^2 \to \mathbb{R}^2$, defined by $P(\alpha) = u_\alpha(T)$. Under suitable conditions, an application of Brouwer's Theorem ensures the existence of a solution to the periodic problem.

Our main result is the following

Theorem 1.2. *Let μ be a finite positive Borel measure on $[0, T]$. Let $\Omega \supset [0, T] \times \mathbb{R}^n \to \mathbb{R}^n$ be an open set. Let $f : \Omega \to \mathbb{R}^n$ be a function satisfying the following conditions:*

- (H_1) *For each $x \in \mathbb{R}^n$, $f(t, x)$ is μ-measurable with respect to the variable t.*
- (H_2) *The function f is Lipschitz with respect to the variable x.*
- (H_3) *f is continuous with respect to t at every point such that $\mu(\{t\}) = 0$.*
- (H_4) *There exists a ball $B(0, R)$ such that:*
  - (H_{41}) *There exists $x^* \in B(0, R)$ such that $f(\cdot, x^*) \in L^2([0, T], d\mu)$.*
  - (H_{42}) *For every $t \in [0, T]$ and every $x \in \overline{B(0, R)}$, we have*
  $$x + f(t, x)\mu(\{t\}) \in \overline{B(0, R)}.$$
  - (H_{43}) *For every $t \in [0, T]$ with $\mu(\{t\}) = 0$ and every x such that $|x| = R$, we have $f(t, x) \cdot x < 0$.*

Then, problem (MDE) has at least one solution.

The paper is organized as follows. In Section 2 we review some definitions and results on Lebesgue–Stieltjes measures for functions of bounded variation. In Section 3 we prove a version of Gronwall's lemma involving inequalities with respect to a positive Borel measure instead of the Lebesgue measure. This lemma will be used to prove the main theorem in Section 4. Finally, in Section 5, we apply our result to the problem of finding periodic solutions to a system describing a pendulum subject to an impulsive force. We also present a computational implementation of the shooting method to approximate the corresponding solution.

2. Preliminaries

As usual, we denote by $\mathbb{R}^n$ the set of all n-tuples with real components. If $x \in \mathbb{R}^n$, we write the Euclidean norm as $|x| = \sqrt{x_1^2 + \cdots x_n^2}$, and the taxicab norm as $|x|_1 = |x_1| + \cdots + |x_n|$. We use the dot product on $\mathbb{R}^n$ defined by $x \cdot y = x_1 y_1 + \cdots + x_n y_n$.

The set of all continuous functions $f : [0,T] \to \mathbb{R}^n$ will be denoted by $C([0,T],\mathbb{R}^n)$, or simply $C([0,T])$ when the codomain is $\mathbb{R}$.

We briefly introduce some notation related to measure theory; we refer to [7,10,11,15,24] for further details on the material presented in this section.

Given a Borel measurable subset A of $\mathbb{R}$, we denote by $\mathscr{B}(A)$ the σ-algebra of all Borel subsets of A. If μ is a signed measure, we write $\mu = \mu^+ - \mu^-$ for the Jordan decomposition of μ, and $|\mu|$ for the *total variation* of μ, that is, $|\mu| = \mu^+ + \mu^-$. We write $\mu \perp \eta$ to indicate that μ and η are *mutually singular* measures. We say that a measure μ is *continuous* if, for every $t \in \Omega$, it holds that $\mu(\{t\}) = 0$. We use the symbol λ to denote the Lebesgue measure on $\mathbb{R}$.

As mentioned above, the set of all functions of bounded variation $\varphi : [0,T] \to \mathbb{R}^n$ is denoted by $BV([0,T],\mathbb{R}^n)$. If $\varphi \in BV([0,T],\mathbb{R}^n)$, we denote by $\mathrm{V}(\varphi,[0,T])$ the *total variation* of φ. As is well known, functions in $BV([0,T],\mathbb{R}^n)$ have one-sided limits at every $t \in [0,T]$ and are continuous except at a set that is at most countable. If $\varphi \in BV([0,T],\mathbb{R}^n)$, then there exists a unique Borel measure, denoted by μ_φ, such that $\mu_\varphi([a,b)) = \varphi(b^-) - \varphi(a^-)$ for every a and b in $[0,T]$. This measure is called the *Lebesgue–Stieltjes measure induced by* φ (see [11] for details). We will usually assume that φ is left-continuous, in which case $\mu_\varphi([a,b)) = \varphi(b) - \varphi(a)$.

The existence of local and global solutions to initial value problems for MDEs has been studied extensively in the literature. We refer, in particular, to the approach developed in [4], where the following theorem is established. We adapt its statement to suit the needs of this article.

Theorem 2.1. *Let μ be a finite measure, Ω be an open set of $\mathbb{R} \times \mathbb{R}^n$, and $(0,\alpha) \in \Omega$. Suppose that $f : \Omega \to \mathbb{R}^n$ is locally Lipschitz with respect to the second variable for $t \geqslant 0$, and that φ is a solution of (IVP) defined on a maximal half-open interval $[0,t_1)$. Then exactly one of the following assertions holds:*

(1) For every $K \Subset \Omega$ there exists $t_2 \in [0,t_1)$ such that $(t,\varphi(t)) \notin K$ for all $t \in (t_2,t_1)$,

(2) The limit $x_1 := \lim_{t \to t_1^-} \varphi(t)$ exists, $(t_1,x_1) \in \Omega$, and $(t_1, x_1 + f(t_1,x_1)\mu(\{t_1\})) \notin \Omega$.

If $|\mu|(\{t_1\}) = 0$, then alternative 1 holds. Otherwise, either of the two assertions may occur.

3. A Gronwall-Type Lemma for Measures

We first prove a Gronwall-type inequality for continuous measures and then extend this result to arbitrary positive Borel measures.

Lemma 3.1. *Let μ be a finite, positive, continuous measure. If $u \in L^1(\mu)$ satisfies*

$$u(t) \leqslant c + \int_{[0,t)} u(r)\, d\mu, \tag{3.1}$$

then $u(t) \leqslant c e^{\mu([0,t))}$.

Proof. We define $w(t) = c + \int_{[0,t)} u(r)\ d\mu(r)$, $F(s) = -e^{-s}$, and $h(t) = \mu([0,t))$. Note that h is a continuous function and $\mu_h = \mu$. The Lebesgue–Stieltjes measure μ_w associated with w satisfies $\mu_w \ll \mu$ and $d\mu_w = u d\mu$. Additionally, using the chain rule for BV functions (see [1, Th. 3.99]; note that μ has no jump part), we have that $\mu_{F\circ h} = F'(h(r))\,\mu$. Therefore, we obtain

$$\int_{[0,t)} -F(h(r))\ dw \leqslant \int_{[0,t)} F'(h(r))\, w(r)\ d\mu = \int_{[0,t)} w(r)\ d\mu_{F\circ h}.$$

Then, using the integration by parts formula (see [24, Th. 14.1]) and taking into account the continuity of $F \circ h$, we have

$$\begin{aligned} 0 &\leqslant \int_{[0,t)} F(h(r))\ dw + \int_{[0,t)} w(r)\ \mu_{F\circ h} = \mu_{(F\circ h)w}([0,t)) \\ &= F(h(t))w(t) - F(h(0))w(0). \end{aligned}$$

Therefore, we conclude

$$u(t) \leqslant w(t) \leqslant c \exp\left(\mu([0,t))\right).$$

□

Definition 3.2. *Let μ be a finite positive measure and let $D = \{\tau \in [0,T] \mid \mu(\{\tau\}) > 0\}$.*

For any Borel set A, we define the jump part of μ *as the measure μ_j induced by μ on D, that is,*

$$\mu_j(A) := \sum_{\tau \in D\cap A} \mu(\{\tau\}) = \mu\left(D \cap A\right).$$

Moreover, we introduce the continuous measure

$$\bar{\mu}\left(A\right) := \mu(A) - \mu_j\left(A\right) = \mu\left(A \backslash D\right).$$

In other words, in the terminology of Ambrosio et al. (see [1]), $\bar{\mu}$ is the sum of the Cantor and absolutely continuous parts of μ. Note that μ_j and $\bar{\mu}$ are mutually singular Borel measures.

Theorem 3.3 (**Generalized Gronwall Inequality**). *Let μ be a finite, positive measure. If $u \in L^1(\mu)$, $u \geqslant 0$ and it satisfies*

$$u(t) \leqslant c + \int_{[0,t)} u(r)\ d\mu,$$

then

$$u(t) \leqslant c e^{K(t)\mu([0,t))},$$

where $K(t) = \prod_{\tau \in D\cap(0,t)} (1 + \mu(\{\tau\}))$.

Proof. We fix $\epsilon > 0$. By the absolute continuity of the integral, there exists $\delta > 0$ such that

$$\text{if } \mu(A) \leqslant \delta \quad \text{then} \quad \left|\int_A u(r)\ d\mu\right| \leqslant \epsilon. \tag{3.2}$$

We define $w(t) = c + \int_{[0,t)} u(s)\, d\mu$. Fix an arbitrary $t \in [0,T)$. Since D is countable, we may assume that $D \cap [0,t) = \{t_n\}_{n=1}^{\infty}$. In addition, since

$$\sum_{n=1}^{\infty} \mu(\{t_n\}) = \mu(D \cap [0,t)) < \infty, , \tag{3.3}$$

there exists $N \in \mathbb{N}$ such that

$$\sum_{n=N+1}^{\infty} \mu(\{t_n\}) \leqslant \delta. \tag{3.4}$$

Let $\{s_1, \dots, s_N\}$ be an increasing reordering of the set $D_\delta := \{t_1, \dots, t_N\}$. Additionally, we set $s_{N+1} = t$.

Decomposing the integral and using the inequality $u \leqslant w$, we obtain

$$\begin{aligned} w(s_{j+1}) &\leqslant w(s_j)\,[1 + \mu(\{s_j\})] + \int_{(s_j, s_{j+1})} u(s)\, d\mu, \\ &\leqslant w(s_j)\,[1 + \mu(\{s_j\})] + K(t) \int_{(s_j, s_{j+1})} u(s)\, d\mu \end{aligned} \tag{3.5}$$

for $j = 1, \dots, N$.

By an inductive argument, and then using that $\mu = \bar{\mu}$ on $[0,t)\backslash D$, (3.2), (3.4) and finally that $\bar{\mu}(D) = 0$, we obtain

$$\begin{aligned} w(t) &\leqslant cK(t) + K(t) \int_{[0,t) - \{t_1, \cdots, t_N\}} u(s)\, d\mu \\ &= cK(t) + K(t) \int_{[0,t)\backslash D} u(s)\, d\mu + K(t) \int_{D\backslash\{t_1, \cdots, t_N\}} u(s)\, d\mu \\ &\leqslant cK(t) + K(t) \int_{[0,t)\backslash D} u(s) d\,\bar{\mu} + K(t)\epsilon \\ &= cK(t) + K(t)\epsilon + K(t) \int_{[0,t)} u(s)\, d\bar{\mu} + \epsilon. \end{aligned}$$

Doing $\epsilon \to 0$ and taking account of that $K(r)$ is increasing for every $r \in [0,t]$, we get

$$u(r) \leqslant w(r) \leqslant cK(t) + K(t) \int_{[0,r)} u(s)\, d\bar{\mu}.$$

Now, since $\bar{\mu}$ is a continuous measure, we are in a position to apply Lemma 3.1. We obtain

$$u(t) \leqslant cK(t) e^{K(t)\bar{\mu}([0,t))}.$$

On the other hand,

$$K(t) \leqslant \exp\left(\sum_{s \in D \cap [0,t)} \mu(\{s\}) \right) \leqslant \exp\left(K(t)\mu_a\left([0,t)\right)\right).$$

The statement follows from the last inequalities. □

4. Proof of Main Theorem

Throughout this section, we will assume that f is as in the statement of Theorem 1.2.

In what follows, when considering solutions of (IVP), we will assume that they are defined on a maximum half-open interval $I_\varphi = [0, t_\varphi)$. We note that if $\lim_{t \nearrow t_\varphi} \varphi(t)$ exists, then the left-continuous extension of φ to the closed interval $[0, T]$ is also a solution in that interval.

Proposition 4.1. *Let φ_α be a solution of the (IVP). Then $[0, T] \subset [0, t_\varphi]$ and $\varphi_\alpha \in L^\infty([0, T], \mathbb{R}^n)$. Moreover, there exists a function $0 \leqslant b \in L^2([0, T], d\mu)$ such that if $s \leqslant t$ then*

$$|\varphi_\alpha(t) - \varphi_\alpha(s)| \leqslant \int_{[s,t)} b(r) d\mu(r).$$

If μ is continuous at $t_0 \in [0, T]$, then φ_α is continuous at t_0.

Proof. As a consequence of (H_2) and (H_{41}), we obtain

$$|f(t, x)| \leqslant |f(t, x^*)| + a|x - x^*| = \gamma(t) + L|x|, \tag{4.1}$$

where L is the Lipschitz constant of f and $\gamma = b + L|x^*| \in L^2([0, t], \mu)$. Therefore, using (1.2), we get

$$|\varphi_\alpha(t)| \leqslant C + L \int_{[0,t)} |\varphi_\alpha(s)| \, d\mu.$$

where $C = |\varphi_\alpha(0)| + \|\gamma\|_{L^1(\mu)}$. Applying Theorem 3.3, deduce

$$|\varphi_\alpha(t)| \leqslant C \exp\left(K(t) L\mu([0, t)\right) \leqslant C \exp\left(K(T) L\mu([0, T])\right), \quad t \in [0, t_\varphi).$$

We conclude that $\varphi_\alpha \in L^\infty([0, t_\varphi), \mathbb{R}^n)$.

If $t_\varphi < T$ were true, statement 1 of Theorem 2.1 would be false because $\varphi_\alpha(t)$ would be inside the compact set $[0, T] \times \overline{B(0, \|\varphi_\alpha\|_{L^\infty})}$ for every $t \in [0, t_\varphi)$. Since alternative 2 of Theorem 2.1 is trivially false, there is nothing left but $t_\varphi \geqslant T$.

On the other hand, for $s \leqslant t$ it holds that

$$\begin{aligned} |\varphi_\alpha(t) - \varphi_\alpha(s)| &\leqslant \int_{[s,t)} |f(r, \varphi_\alpha(r))| \, d\mu \\ &\leqslant \int_{[s,t)} L\|\varphi_\alpha\|_\infty + \gamma \, d\mu =: \int_{[s,t)} b(r) \, d\mu(r). \end{aligned}$$

Note that $b = L\|\varphi_\alpha\|_\infty + \gamma \in L^2([0, T], d\mu)$. The last part of the proposition is an immediate consequence of inequality (4.1). □

Definition 4.2. *We will call the mapping $P : \mathbb{R}^n \to \mathbb{R}^n$ defined by $P(\alpha) = \varphi_\alpha(T)$ Poincaré operator, where φ_α is a solution of (IVP).*

Solutions to initial value problems associated with MDE may be discontinuous with respect to time. The following result shows that such solutions are continuous with respect to the initial condition.

Lemma 4.3. *The Poincaré operator is Lipschitz continuous.*

Proof. Let φ_α and φ_β be two solutions of (IVP). As f is Lipschitz, we have

$$\begin{aligned}|\varphi_\alpha(t)-\varphi_\beta(t)| &\leqslant |\alpha-\beta| + \int_{[0,t)} |f(r,\varphi_\alpha(r)) - f(t,\varphi_\beta(t))|\, d\mu \\ &\leqslant |\alpha-\beta| + L\int_{[0,t)} |\varphi_\alpha(r)-\varphi_\beta(t)|\, d\mu.\end{aligned}$$

Applying Theorem 3.3, we obtain

$$|P(\alpha)-P(\beta)| = |\varphi_\alpha(T)-\varphi_\beta(T)| \leqslant |\alpha-\beta| e^{K(T)L\mu([0,T))}.$$

This inequality proves the assertion of the lemma. □

Recall the properties of radius R from the statement of Theorem (1.2).

Theorem 4.4. *The closed ball $\overline{B(0,R)}$ is an invariant set for the Poincaré map, i.e. $P\left(\overline{B(0,R)}\right) \subset \overline{B(0,R)}$.*

Proof. Let $\alpha \in \overline{B(0,R)}$ and let φ_α be the solution of (IVP) defined on a maximal interval. We define the set

$$A = \left\{ t \in [0,T] \mid \varphi_\alpha(s) \in \overline{B(0,R)},\ \forall s \in [0,t] \right\}.$$

We will see that the set A is simultaneously relatively open and closed to the interval $[0,T]$. Since $[0,T]$ is connected, then $A = [0,T]$.

First, we will prove that A is closed. Let t_n be a sequence in A that converges to t. If $t_n \geqslant t$ for some n, then $\varphi_\alpha(s) \in \overline{B(0,R)}$ for every $s \in [0,t]$, and therefore $t \in A$. Conversely, if $t_n < t$ for every n, since φ_α is left-continuous, we have $\lim_{n\to\infty} \varphi_\alpha(t_n) = \varphi_\alpha(t)$; and, due to $\varphi_\alpha(t_n) \in \overline{B(0,R)}$, it follows straightforwardly that $\varphi_\alpha(t) \in \overline{B(0,R)}$, i.e. $t \in A$. Thus, A is closed.

We observe that by definition A is an interval. If we suppose that A is not open relative to $[0,T]$, then there would be $t_0 \in A$ such that for every $t > t_0$ it holds that $t \notin A$. As $t_0 \in A$ then $\varphi_\alpha(t_0) \leqslant R$. We can consider three cases:

Case 1. Suppose that $\varphi_\alpha(t_0) \in B(0,R)$ and $\mu(\{t_0\}) = 0$. Then by Proposition 4.1, φ_α is continuous at t_0, hence there exists an open interval $[t_0, t_0+\delta)$ such that $\varphi_\alpha([t_0,t_0+\delta)) \subset B(0,R)$, which is a contradiction with the fact that t_0 is the supremum of A.

Case 2. Suppose that $\varphi(t_0) \in \partial B(0,R)$ and $\mu(\{t_0\}) = 0$. According to (H_{43}), $\beta := -f(t_0,\varphi(t_0)) \cdot \varphi(t_0)/|\varphi_\alpha(t_0)| > 0$. By Proposition 4.1 and by (H_3), there exists $\delta > 0$ such that if $t \in [t_0, t_0+\delta)$ then

$$|\varphi_\alpha(s)-\varphi_\alpha(t_0)| < \frac{\beta}{3L} \quad \text{and} \quad |f(s,\varphi_\alpha(t_0)) - f(t_0,\varphi_\alpha(t_0))| \leqslant \frac{\beta}{3}.$$

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$$\begin{aligned}[\varphi_\alpha(t)-\varphi_\alpha(t_0)]\cdot\frac{\varphi_\alpha(t_0)}{|\varphi_\alpha(t_0)|} &\leqslant \int_{[t_0,t)} |f(s,\varphi_\alpha(s))-f(s,\varphi_\alpha(t_0))|\ d\mu \\ &+ \int_{[t_0,t)} |f(s,\varphi_\alpha(t_0))-f(t_0,\varphi_\alpha(t_0))|\ d\mu \\ &+ \int_{[t_0,t)} f(t_0,\varphi_\alpha(t_0))\cdot\frac{\varphi_\alpha(t_0)}{|\varphi_\alpha(t_0)|}\ d\mu \leqslant -\frac{\beta}{3}\mu([t_0,t)).\end{aligned}$$

Considering the first-order Taylor polynomial approximation around $|\varphi_\alpha(t_0)|$ to the Euclidean norm function, taking into account Proposition 4.1, the Cauchy-Schwarz inequality, and the fact that $b \in L^2$,we have

$$\begin{aligned}|\varphi_\alpha(t)|-R &= [\varphi_\alpha(t)-\varphi_\alpha(t_0)]\cdot\frac{\varphi_\alpha(t_0)}{|\varphi_\alpha(t_0)|} + \mathcal{O}\left(\int_{[t_0,t)} b(s)\ d\mu\right)^2 \\ &\leqslant -\frac{\beta}{3}\mu([t_0,t)) + \mathcal{O}\left(\mu([t_0,t))\int_{[t_0,t)} b^2(s)\ d\mu\right)\end{aligned}$$

Since $\lim_{t\searrow t_0}\mu([t_0,t)) = \mu(\{t_0\}) = 0$, the last expression in the previous inequality is negative for t close enough to t_0. Therefore, there exists a neighborhood V of t_0 such that $|\varphi_\alpha(t)| \leqslant R$ for all $t \in V$, i.e. A is an open set. This completes the proof of this case.

Case 3. $\mu(\{t_0\}) \neq 0$. Using the Lebesgue Dominated Convergence Theorem and (H_{42}), we obtain that

$$\lim_{t\searrow t_0}\varphi_\alpha(t) = \varphi_\alpha(t_0) + f(t_0,\varphi_\alpha(t_0))\mu(\{t_0\}) \in B(0,R).$$

Therefore, there exists $\delta > 0$ such that $|\varphi_\alpha(t)| < R$ for all $t \in [t_0, t_0+\delta)$, contradicting the fact that t_0 is the supremum of A.

Then $A = [0,T]$. Consequently, for any $\alpha \in \overline{B(0,R)}$, it holds that $P(\alpha) = \varphi_\alpha(T) \in \overline{B(0,R)}$. □

We recall the well-known Brouwer Theorem, see [12].

Theorem 4.5 (**Brouwer Theorem**). *Let $B(0,R)$ be a ball of $\mathbb{R}^n$ and let $P : \overline{B(0,R)} \to \overline{B(0,R)}$ be a function. If P is continuous, there exists $x \in \overline{B(0,R)}$ such that $P(x) = x$.*

Now, Theorem 1.2 follows immediately.

Proof Theorem 1.2. From Lemma 4.3, the operator P is continuous and by Theorem 4.4, we have $P\left(\overline{B(0,R)}\right) \subset \overline{B(0,R)}$. Now, applying Brouwer Theorem 4.5, there exists a solution φ to the problem (1.1) with $\varphi(T) = \varphi(0)$. □

Remark 4.6. Clearly, we can replace the ball in Theorem 1.2 by any set that satisfies Brouwer's Theorem, in particular any compact and convex set K. In that case, we must to replace the vector x in inequality $f \cdot x < 0$ by the exterior normal vector $\vec{n}$ to K at x. Simultaneously, we can consider the case in which ∂K has subsets invariant under the flow. With all this in mind, we can replace hypothesis (H_{43}) by

(H'_{43}) $\partial K = \Gamma_1 \cup \Gamma_2$ where f satisfies $f(t,x)\cdot\vec{n} < 0$, for every (t,x) with $x \in \Gamma_1$ and $\mu(\{t\}) = 0$, and the set Γ_2 is invariant for flow. That is, if $\mu(t) = 0$ and $\alpha \in \Gamma_2$ then $\varphi_\alpha(s) \in \Gamma_2$ for every $s \geqslant t$.

SA: Un ejemplo??? Al cambiar las condiciones parencen 2...

5. An Example: Impulsively Forced and Damped Pendulum

In order to illustrate the application of our main theorem, we will present a concrete example. We consider the following 2π-periodic boundary value problem

$$\begin{cases} d\theta' = (-\sin\theta - c\theta')\,d\lambda + g(t,\theta,\theta')\,d\mu, \\ \theta(0) - \theta(2\pi) = \theta'(0) - \theta'(2\pi) = 0. \end{cases} \tag{5.1}$$

The equation corresponds to a damped and impulsively forced pendulum with viscosity constant equal to c. We assume that the external force is impulsive and is given by a sum of delta measures

$$\mu = \sum_{i=1}^{n} \alpha_i \delta_{t_i},$$

where $D = \{t_1, \dots, t_n\}$ is a finite subset of $[0, 2\pi]$. We suppose that $\alpha_i \in \mathbb{R}$, i.e. μ can be a signed measure. The function g is supposed to be sufficiently regular. Recall that λ is the Lebesgue measure.

We can write the equation in the problem (5.1) as

$$dx = F(t,x)\,d\nu,$$

where ν is the positive measure $\nu = \lambda + \mu$, $x = (x_1, x_2) = (\theta, \theta')$ and

$$F(t,x) = \chi_{[0,T]\setminus D}(t)\begin{pmatrix} x_2 \\ -\sin x_1 - cx_2 \end{pmatrix} + \chi_D(t)\begin{pmatrix} 0 \\ g(t,x) \end{pmatrix}.$$

Here χ_A denotes the indicator function with respect to the set A.

We will take the set K as

$$K = \left\{ x \mid -\pi \leqslant x_1 \leqslant \pi,\ \frac{x_2^2}{2} - \cos x_1 \leqslant 1 \right\},$$

In physical terms, K is the set of states with energy less or equal than 1, which is a closed and convex set.

Let us show that we are within the scope of the Remark 4.6. We take $\vec{n} = (\sin x_1, x_2)$ and $t \in [0, 2\pi]$ with $\nu(\{t\}) = 0$, i.e. $t \notin D$. We first consider the damped case, that is $c > 0$. In this situation, inequality $F \cdot \vec{n} < 0$ holds when $x \in \Gamma_1 := \partial K \setminus \{(\pm\pi, 0)\}$. The set $\Gamma_2 = \{(\pm\pi, 0)\}$ is trivially invariant for flow, because it consists of equilibrium points. In the undamped case ($c = 0$), the set ∂K is invariant. This is an immediate consequence of an energy preserving argument and the fact that $c = 0$ and $g = 0$ on ∂K. In conclusion, the hypothesis in Remark 4.6 is satisfied.

In order for F to satisfy (H_{42}) we have to choose g appropriately. For example, if g satisfies that $|g(t,x)| \leqslant d(x, \partial K)/M$, where $M = \sup_{t\in D}\{\nu(\{t\})\}$, then ($H_{42}$) is valid. Under all of the above assumptions, problem (5.1) has a solution.

The shooting method can be used as an algorithm to numerically solve periodic problems associated to MDEs. We programmed this algorithm using Python language [1]. It numerically calculates Lebesgue-Stieltjes integrals and solves MDEs using Picard's successive approximation method. The convergence of this last procedure was established in [4]. This is used to obtain the Poincaré mapping P. We then plot the contour lines and density plots of $h(x) = \|x - P(x)\|^2$. This type of graph gives us clues about the region where it is convenient to look for a fixed point of P, and also potentially shows us all these fixed points. Finally, the fixed point problem $P(x) = x$ is solved using optimizations functions of the modulus `scipy.optimize`.

Successive approximations are perhaps not the most efficient algorithm in solving MDEs; our plan is to explore the issue of numerical solutions for MDE in greater depth in a future article.

Returning to the particular problem (5.1), we consider two experiments that are subject to all of the above hypotheses, and thus our Theorem 1.2 guarantees the existence of periodic solutions. We are interested in finding these solutions numerically and interpreting them in physical terms.

Experiment 1: damped pendulum. We take $c = 2.0$, $D = \{1, \dots, 5\}$, $\mu = \sum_{j=1}^{5}(-1)^j \delta_j$ and $g(t, x) = \max\{1.9 - \|x\|, 0\}$. Then $M = 1$, and it is not hard to see that g verifies the inequality $|g| \leqslant d(x, \partial K)$. In Figure 3 the density plot of $h(x) = \|x - P(x)\|^2$ is drawn.

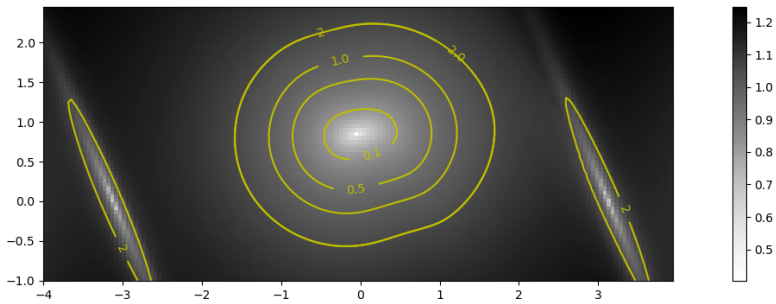

FIGURE 1. Level sets of $h(x) = \|x - P(x)\|^2$, $c = 2.0$, $\mu = \sum_{j=1}^{5}(-1)^j \delta_j$.

The density map presumably shows that there are only three periodic solutions. Two of them correspond to the unstable equilibria $x = (\pm\pi, 0)$ of the simple pendulum, which continue to be equilibria of our equation. Note that the stable equilibrium $(0, 0)$ of the simple pendulum is no longer an equilibrium of our equation, since the impulsive forces alter its steady state. This is not true for unstable equilibria, since $g = 0$ in ∂K. A third solution can be seen in the center of the image. Its position $(-0.04, 0.85)$ was determined more accurately using the optimization algorithm. In Figure 2, we plot the corresponding solution. sAs can be observed, in this case, the mass spends most of its motion at negative angles. This is because the initial impulse is negative; if it were positive, the mass would spend most of its time at positive angles.

: **LS:**programe la solución y efectivamente si $\mu = \sum_{j=1}^{5}(-1)^{j+1}\delta_j$ la solución se refleja con respecto al eje x.

[1] https://github.com/fdmazzone/shootingMDE

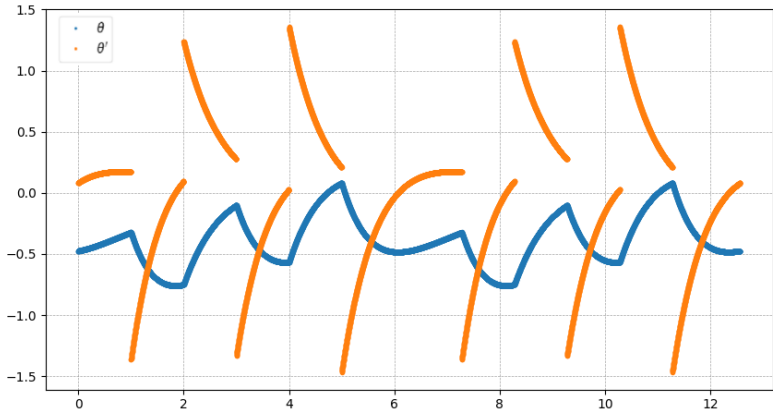

FIGURE 2. Graph solution (θ, θ') of (1.1) in the interval $[0, 4\pi]$ with $c = 2.0$ and $\mu = \sum_{j=1}^{5}(-1)^j \delta_j$.

Experiment 2: undamped pendulum. In this example, we take $c = 0.0$, $D = \{1, \ldots, 5\}$ (again) and $\mu = 0.5 \sum_{j=1}^{5} \delta_j$. The function g is the same as in the previous example. In this case, we are interested in seeing what a periodic solution looks like when the impulsive forces always act in the same direction. In Figure 3 the density plot of $h(x) = \|x - P(x)\|^2$ is drawn. The map clearly shows the compact region K and the solutions to the equation $h(x) = 0$. The unstable solutions $(\pi, 0)$ and $(-\pi, 0)$ lie on the border of K, where the pendulum behaves like a simple pendulum because the external force g vanishes on the border of K. As in the previous example, the stable equilibrium $(0, 0)$ of the simple pendulum is no longer an equilibrium of our equation. We observe a solution of $h(x) = 0$ inside K, at the point $P = (1.338, 0.578)$, which corresponds to the periodic solution we were looking for, obtained through an optimization algorithm. Its graph is shown in Figure 4.

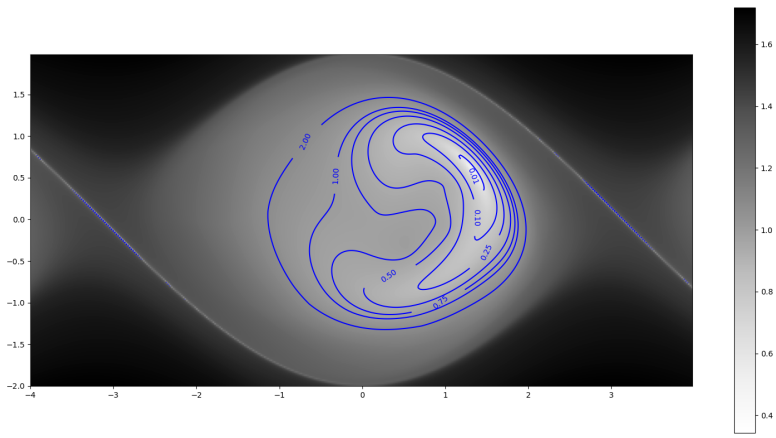

FIGURE 3. Level sets of $h(x) = \|x - P(x)\|^2$, $c = 0.0$, $\mu = 0.5 \sum_{j=1}^{5} \delta_j$.

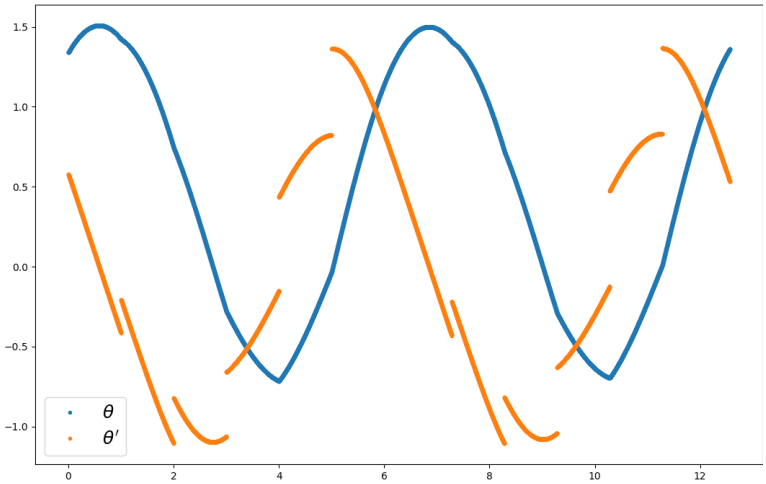

FIGURE 4. Graph solution (θ, θ') of (1.1) in the interval $[0, 4\pi]$ with $c = 0.0$ and $\mu = 0.5 \sum_{j=1}^{5} \delta_j$.

REFERENCES

[1] L. Ambrosio, N. Fusco, and D. Pallara. *Functions of Bounded Variation and Free Discontinuity Problems*. Clarendon Press, 2000.

[2] D. Bainov and P. Simeonov. *Impulsive Differential Equations: Periodic Solutions and Applications*. CRC Press, jul 1993.

[3] D. D. Bainov and P. S. Simeonov. *Impulsive Differential Equations: Asymptotic Properties of the Solutions*. World Scientific, mar 1995.

[4] G. Beltritti, S. Demaria, G. Giubergia, and F. Mazzone. *The Picard-Lindelöf theorem and continuation of solutions for measure differential equations*. Czechoslovak Mathematical Journal, may 2023.

[5] E. M. Bonotto, M. Federson, and J. G. Mesquita. *Generalized Ordinary Differential Equations in Abstract Spaces and Applications*. wiley, 2021.

[6] B. Brogliato. *Nonsmooth Mechanics: Models, Dynamics and Control*. Springer, dic 100.

[7] M. Carter. *The Lebesgue-Stieltjes Integral: A Practical Introduction*. Springer, dic 2012.

[8] N. Daoudi-Merzagui and A. Boucherif. Variational approach to impulsive differential equations with singular nonlinearities. *Journal of Applied Mathematics*, 2013, 2013.

[9] J. Diestel and J. J. Uhl. *Vector Measures*. American Mathematical Soc., jun 1977.

[10] L. C. Evans and R. F. Gariepy. *Measure Theory and Fine Properties of Functions, Revised Edition*. Chapman and Hall/CRC, ene 2018.

[11] G. B. Folland. *Real Analysis: Modern Techniques and Their Applications*. Wiley, may 2007.

[12] A. Fonda. *Playing Around Resonance: An Invitation to the Search of Periodic Solutions for Second Order Ordinary Differential Equations*. Springer International Publishing, 2016.

[13] J. R. Graef, S. Heidarkhani, and L. Kong. Nontrivial periodic solutions to second-order impulsive hamiltonian systems. *Electron J. Differ. Equ*, 2015(204):1–17, 2015.

[14] S. Heidarkhani, M. Ferrara, and A. Salari. Infinitely many periodic solutions for a class of perturbed second-order differential equations with impulses. *Acta Applicandae Mathematicae*, 139(1):81–94, 2015.

[15] E. Hewitt and K. Stromberg. *Real and Abstract Analysis: A Modern Treatment of the Theory of Functions of a Real Variable*. Springer New York, 1975.
[16] Z. L. Jingli Xie. The existence of solutions of periodic bvp for second order impulsive differential equations. *Journal of Inequalities and Applications*, 2013, 12 2013.
[17] J. Kurzweil. *Generalized Ordinary Differential Equations: Not Absolutely Continuous Solutions*. World Scientific, ene 2012.
[18] V. Lakshmikantham, D. Bainov, and P. Simeonov. *Theory of Impulsive Differential Equations*. Series in modern applied mathematics. World Scientific, 1989.
[19] G. A. Leonov, H. Nijmeijer, A. Y. Pogromsky, and A. L. Fradkov. *Dynamics and Control of Hybrid Mechanical Systems*. World Scientific, ene 2010.
[20] C. Li, L. Li, and H. Yang. Infinitely many solutions for non-autonomous second-order systems with impulsive effects. *Journal of Applied Analysis & Computation*, 10(2):427–441, 2020.
[21] H. Z. Z. Li. Variational approach to impulsive differential equations with periodic boundary conditions. *Nonlinear Analysis: Real World Applications*, 11:67–78, 2010. Métodos Variacionales, impulsivas.
[22] J. Li. Lazer-leach type condition for second order differential equations at resonance with impulsive effects via variational method. *Boundary Value Problems*, 2014:233, 2014. Métodos Variacionales, impulsivas, periodicas.
[23] W. Liu, Jian; Yu. Two solutions to superlinear hamiltonian systems with impulsive effects. *Applied Mathematics Letters*, 11 2019.
[24] S. Saks. *Theory of the Integral*. Dover Publications, 2005.
[25] S. Schwabik. *Generalized Ordinary Differential Equations*. World Scientific, oct 1992.
[26] S. Schwabik, M. Tvrdý, and O. Vejvoda. *Differential and Integral Equations: Boundary Value Problems and Adjoints*. Academia Praha, may 1979.
[27] W. Tian, Yu; Ge. Applications of variational methods to boundary-value problem for impulsive differential equations. *Proceedings of the Edinburgh Mathematical Society*, 51, 2008. Métodos Variacionales, impulsivas.
[28] X. Yang and H. Chen. Periodic solutions for autonomous (q, p)-laplacian system with impulsive effects. *J. Applied Mathematics*, 2011:378389–1, 2011.
[29] S. T. Zavalishchin and A. N. Sesekin. *Dynamic Impulse Systems: Theory and Applications*. Springer, feb 1997.
[30] H. Zhang and Z. Li. Periodic solutions of second-order nonautonomous impulsive differential equations. *International Journal of Qualitative Theory of Differential Equations and Applications*, 2(1):112–124, 2008.

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